Segmentation of Carotid Plaques Based on Improved DeepLabV3

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Abstract

This paper proposes a neural network, MOGA-GCNDeepLabV3+, to improve carotid plaque segmentation in ultrasound images affected by noise, low contrast, and blurred edges. DenseNet replaces the backbone of DeepLabV3+, and MogaBlock from MOGANet enhances feature extraction. Additionally, dynamic convolutional networks (GCN) run in parallel with ASPP's dilated convolution to better model complex plaque shapes. A new loss function, TverskyLDAMLoss, combines Tversky and LDAM losses to address pixel class imbalance. Compared to DeepLabV3+, the improved model achieves 3.08% higher mIoU, 2.42% higher mDice, and 3.42% higher mAcc.

CCS Concepts

• **Applied computing** → Life and medical sciences; Health informatics.

Keywords

carotid plaque segmentation, ultrasound images, DenseNet, dynamic convolution, DeepLabV3+, TverskyLDAMLoss

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1 INTRODUCTION

Cardiovascular disease is a leading cause of death and morbidity worldwide [1, 2]. According to the 2023 World Heart Report, 20.5 million people died from cardiovascular diseases in 2021, accounting for one-third of global deaths. In comparison, cardiovascular deaths were 12.1 million in 1990 and 18.6 million in 2019 [3]. By 2050, the standardized incidence rate is expected to reach 663.6 per 100,000 people [4]. Carotid plaques, an early sign of atherosclerosis, are key predictors of cardiovascular events. Although ultrasound imaging is widely used for plaque detection, traditional manual segmentation methods are subjective and time-consuming, limiting large-scale screening and precise analysis. With the rapid growth of medical imaging data, developing efficient and accurate automatic segmentation algorithms has become essential.

Recent advances in deep learning have improved medical image segmentation. Methods such as ResNet50 and Inception_v3 (with data augmentation and filtering) [5], U-Net [6], and HRU-Net (with a hybrid dilated convolution module) [7, 10] address noise interference, class imbalance, and edge blurring. Ding et al. [8] pretrained models on distorted image pairs to improve plaque contour detection, but this requires significant computational resources. Adaptive ensemble models [9] further enhance segmentation but increase computational costs. Similarly, Transformer-based networks like AFS-TPNet improve accuracy but need optimization for inference speed [11].

Despite these advances, challenges remain in feature extraction, edge modeling, and class imbalance. To address these, this paper proposes MOGA-GCNDeepLabV3+, an enhanced version of DeepLabV3+. It replaces the dense blocks in DenseNet with MOGA blocks to enhance multi-scale feature extraction. Additionally, GCN [12] is integrated into the ASPP module to better capture complex shapes and edges. Finally, TverskyLDAMLoss combines Tversky [14] and LDAM losses [13] to address class imbalance, improving segmentation accuracy for small plaques.

2 RELATED WORK

2.1 Network Architecture for deeplabv3+

DeepLabV3+ is a deep learning model for semantic segmentation, consisting of an encoder and decoder. The encoder uses a backbone network and atrous spatial pyramid pooling (ASPP) with dilation rates of 12, 24, and 32 to capture multi-scale features and expand the receptive field efficiently. The decoder upsamples the encoder's output to restore image resolution, and a 1×1 convolution layer generates the final segmentation result, achieving precise target segmentation. Deeplabv3 + model structure is shown in Figure 1.

2.2 Feature Extraction Network Densenet Network

Although DeepLabV3+ excels in semantic segmentation, its large parameters and slow prediction speed reduce efficiency in carotid plaque segmentation. To improve performance, this paper replaces the DenseNet201 backbone in DeepLabV3+ to better handle low contrast and high noise in ultrasound images. Additionally, the Moga module from MogaNet replaces the Dense block, enhancing multi-scale feature extraction with multi-order depthwise separable

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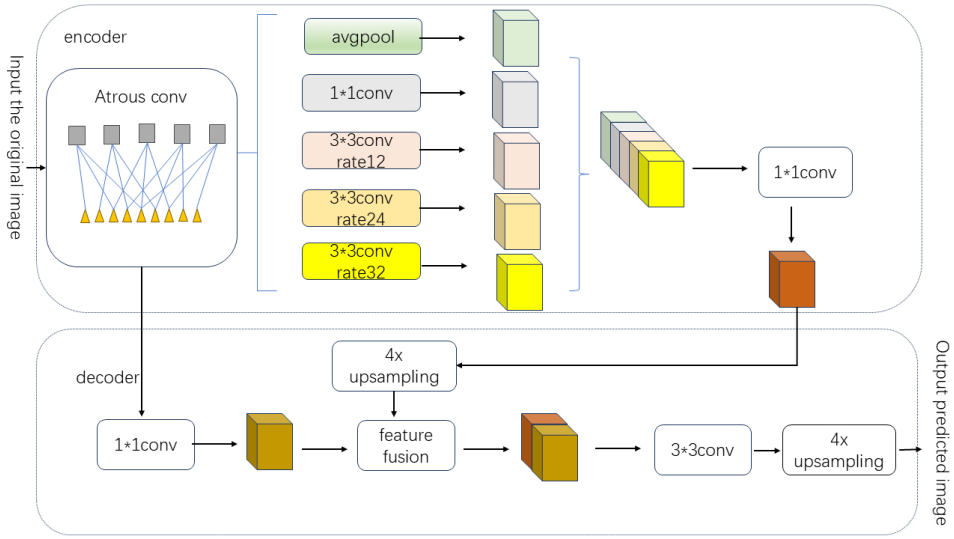

Figure 1: Structure of the deeplabv3+ model.

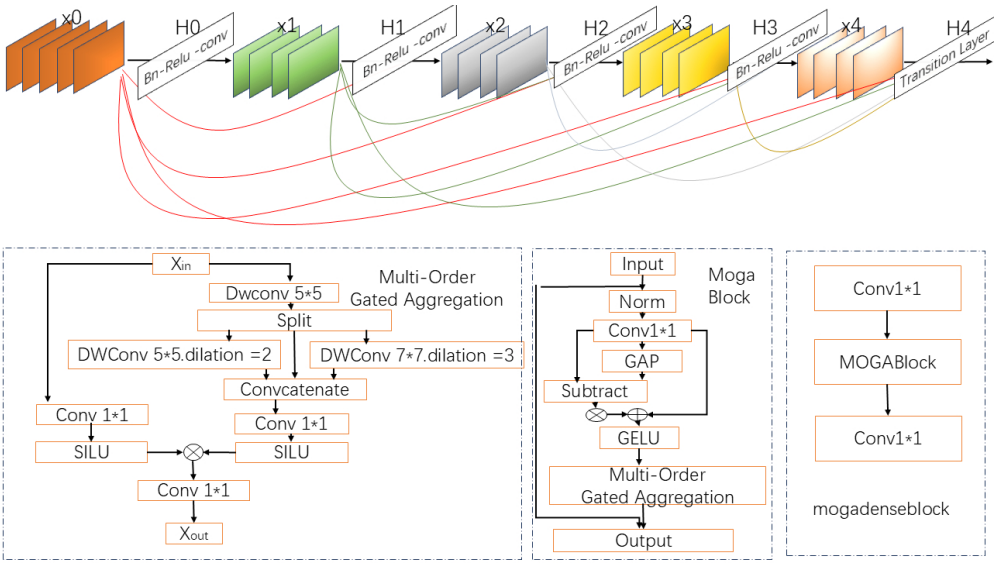

Figure 2: Structure of the densenet model with mogadenseblock modules.

convolutions and a gated aggregation mechanism to emphasize key features and suppress noise. These improvements significantly boost the accuracy and efficiency of ultrasound image segmentation. The structures of DenseNet and MogaDenseBlock are shown in Figure 2.

First, the input feature map $Xin$is compressed using a 1×1 convolution to generate an intermediate feature map A1, reducing computation while preserving key features. The MogaDenseBlock then applies Multi-Order Depthwise Separable Convolutions to capture multi-scale contextual information with different dilation rates, improving performance on complex backgrounds and detailed features. The Gated Aggregation Mechanism adaptivdely emphasizes important features and suppresses redundant information by calculating similarity weights F(i,j), which are transformed into a probability distribution using Softmax and added to A1, producing the enhanced feature map DDD. Finally, 1×1 convolutions and depthwise separable convolutions process the feature map before it is passed to the next layer via DenseNet's connectivity, improving both computational efficiency and accuracy when handling complex shapes and backgrounds.

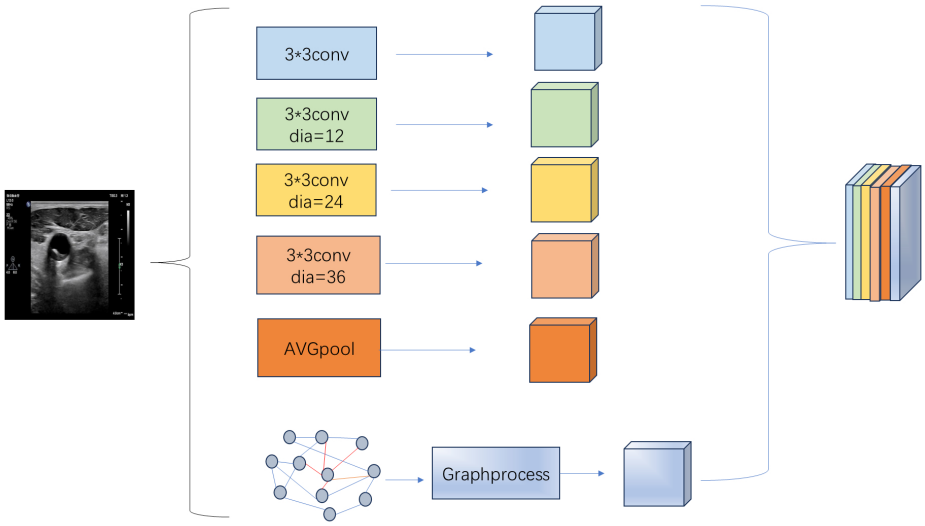

Figure 3: Hybrid module structure of aspp and Grapher.

2.3 Introducing the Grapher module

In medical image segmentation, particularly for complex shapes and ambiguous boundaries like carotid artery plaques, capturing global contextual information is essential. Traditional convolution operations, however, are limited by local receptive fields, making it difficult to manage long-range pixel dependencies. While the ASPP (Atrous Spatial Pyramid Pooling) module captures multi-scale contextual information through atrous convolutions, its focus remains primarily on local regions. To overcome this limitation, this paper integrates the Grapher module into the ASPP process, enhancing the model's ability to capture global features. The structure of the Grapher module is shown in Figure 3.

The ASPP module first extracts multi-scale feature maps using dilated convolutions, which are then passed to the Grapher module to be constructed into a graph structure. In this structure, each node represents a pixel or region, and edges are dynamically generated based on spatial proximity and feature similarity. The Grapher module uses graph convolution to aggregate node features, introducing contextual information from distant nodes and addressing CNNs' limitations in capturing long-range dependencies. The ASPP and Grapher modules collaborate through feature fusion, where ASPP's multi-scale features are integrated with Grapher's global features via element-wise weighted fusion, represented as:

$$Z = M(X) \odot X + (1 - M(X)) \odot Y \tag{1}$$

Here, Z represents the fused feature map, X denotes the local features extracted by the ASPP module, and Y indicates the global features generated by the Grapher module. M(X) is the weight matrix learned through the graph convolutional network, which dynamically adjusts the contribution of local and global features. This fusion mechanism allows the network to flexibly balance between local details and global contextual information, ensuring that dependencies between multi-scale information and distant pixels are fully considered when handling complex boundaries and diverse morphological targets.

2.4 hybrid loss function

In this context, Z represents the fused feature map, X denotes the local features extracted by the ASPP module, and Y indicates the global features generated by the Grapher module. The weight matrix M(X), learned through the graph convolutional network, dynamically adjusts the balance between local and global feature contributions. This fusion mechanism allows the network to flexibly transition between local details and global context, ensuring that multi-scale dependencies and distant pixel relationships are fully considered when handling complex boundaries and morphologically diverse targets.

$$L = \alpha \cdot L_{\text{Tversky}} + \beta \cdot L_{\text{LDAM}} \tag{2}$$

In this context, α and β are weight parameters used to balance the Tversky loss and LDAM loss. Experiments demonstrate that the composite loss function achieves optimal performance when α =0.3 and β =0.7.. In this configuration, LDAM loss is more sensitive to small target categories (such as small area patches), while Tversky loss adjusts the balance between false positives (FP) and false negatives (FN) to improve segmentation results. The Tversky loss is expressed as follows:

$$L_{\text{Tversky}} = 1 - \frac{TP + \epsilon}{TP + \alpha_t \cdot FP + \beta_t \cdot FN + \epsilon} \tag{3}$$

In this context, TP, FP, and FN represent the number of true positives, false positives, and false negatives, respectively. α_t and β_t are hyperparameters that control the weighting of FP and FN, while ϵ is a smoothing term that ensures numerical stability in the calculations. By adjusting α_t and β_t , this loss function can flexibly handle class imbalance and improve segmentation accuracy for small targets. Additionally, the LDAM loss [13] provides boundary adjustments for minority classes (such as small area patches) by

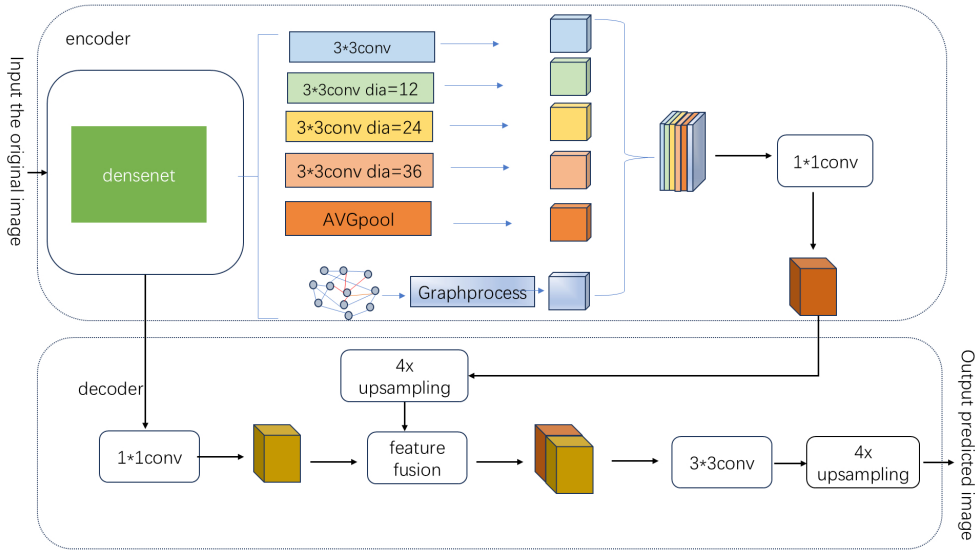

Figure 4: MOGA-GCNDeeplabv3+ model architecture.

dynamically adjusting the decision boundaries between categories. The definition of the LDAM loss function is as follows:

$$L_{\text{LDAM}} = \frac{1}{N}\sum_{i=1}^{N} max\left(0, 1 - \hat{y}_i\left(y_i - \Delta_i\right)\right) \tag{4}$$

where y_i denotes the correct category, $\hat{y}_i$ is the model's prediction score for sample i, $\Delta_i = \frac{C}{n_i^{0.25}}$ is the boundary correction term for the minority category, n_i is the number of samples for category i, and C is a constant to control the strength of the correction. With such a correction mechanism, LDAM effectively improves the sensitivity of the model to a few categories, thus enhancing the segmentation performance of small targets.In summary, TverskyLDAMLoss effectively addresses the class imbalance problem by combining Tversky loss and LDAM loss, and it enhances the detection and segmentation accuracy of small target patches through dynamic weight adjustment.

2.5 Model architecture of the improved deeplabv3+

The architecture of the MOGA-GCNDeepLabV3+ model proposed in this paper is illustrated in Figure 4. In the encoder, DenseNet serves as the backbone to reduce parameter count and improve segmentation efficiency. Unlike traditional DenseNet201, this model replaces the dense block with MogaBlock, enhancing multi-scale feature fusion. MogaBlock automatically allocates weights to features at different levels, optimizing information flow, and excels at handling complex shapes and multi-scale patches, improving small target recognition.

In the ASPP (Atrous Spatial Pyramid Pooling) module, the model introduces parallel use of Dynamic Graph Convolutional Networks (GCN) and dilated convolution, effectively combining local and global information to improve segmentation accuracy—especially useful for carotid plaque regions with blurred boundaries. Additionally, the TverskyLDAMLoss function addresses class imbalance through dynamic weight adjustments, enhancing the segmentation accuracy of small plaques.

In summary, MOGA-GCNDeepLabV3+ significantly improves model efficiency and segmentation performance by integrating DenseNet201, MogaBlock, GCN, and TverskyLDAMLoss.

3 EXPERIMENTS AND ANALYSIS OF RESUITS

3.1 Setting of parameters and evaluation criteria

The environment and parameters of this experiment are shown in Table 1. The training period is 640*640 pixels, the size of the training period is set to 100 epochs, the batch size is 16, and SGD is used as the optimizer with an initial learning rate of 0.001.

This paper evaluates segmentation performance using several metrics: mIoU(mean Intersection over Union), which measures overlap between predicted and ground truth segments; mAcc (mean Accuracy), the average pixel classification accuracy across categories; mDice (mean Dice coefficient), which reflects the similarity between predicted and true results; mPrecision, the average precision across categories, and mRecall, which measures the ability to correctly identify positive samples. These metrics comprehensively assess the model's accuracy and segmentation quality.

$$\text{IoU} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive} + \text{False Negative}} \tag{5}$$

$$\text{Accuracy}_i = \frac{\text{TP}_i}{\text{Total Pixels in Class } i} \tag{6}$$

$$\text{Dice} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}} \tag{7}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

Table 1: Experimental environment.

Parameter	Configuration
Operating System	Window11.64
CPU	Intel(R) Xeon(R) Gold 6136 CPU @ 3.00GHz 2.99 GHz
GPU	NVIDIA GeForce RTX 3090
RAM	32G
CUDA	CUDA 12.1
Deeplearning Framework	PyTorch
Programming language	Python 3.8

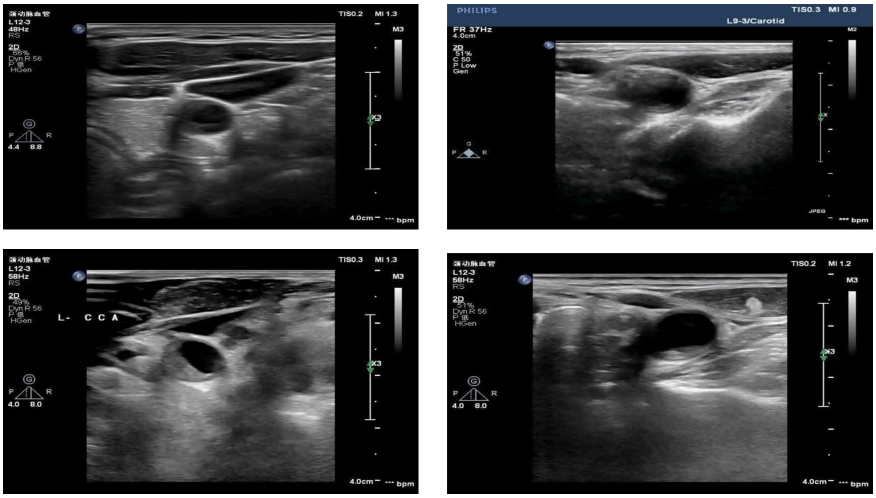

Figure 5: Ultrasound carotid plaque images at different location.

Table 2: DGT-deeplabv3+ network ablation results.

DeepLabV3+	densenet	Grapher	TverskyLDAMLoss	mIoU	mAcc	mDice	mPrecision	mRecall
+				78.01	82.67	85.96	90.01	82.67
+	+			78.53	83.2	86.38	90.25	83.2
+	+		+	80.91	86.2	88.25	90.54	86.20
+	+	+	+	81.09	86.09	88.38	91	86.09

3.2 Data sets and data enhancement

All the data used in this experiment were provided by Inner Mongolia Affiliated Hospital, which were expanded to 1440 images after translation transformation, rotation transformation and other image enhancement methods, with a training set of 1200 and a test set of 240. Figure 5 shows different ultrasound carotid artery plaque images.

3.3 Ablation experiment

The ablation experiment results are shown in Table 2. Using the basic DeepLabV3+ model, the mIoU, mAcc, and mDice are 78.01, 82.67, and 85.96, respectively. Replacing the backbone with DenseNet201 and introducing MogaBlock for multi-scale feature fusion improved these metrics to 78.53, 83.20, and 86.38. Adding the Grapher module further increased mIoU to 80.91, mAcc to 86.20, and mDice to 88.25, showing its effectiveness in handling complex boundaries. Finally, applying the TverskyLDAMLoss function improved mIoU, mAcc, and mDice to 81.09, 86.09, and 88.38, significantly enhancing small target detection and segmentation accuracy.

3.4 Comparison of segmentation effect with other models

To evaluate the segmentation performance of the proposed DGT-DeepLabV3+ model, we conducted comparison experiments with several popular semantic segmentation models, including DeepLabV3+, PSPNet, Segformer, BiSeNetv2, FPN, and GCN. The models were assessed using metrics like mIoU, mAcc, mDice, mPrecision, and mRecall for a comprehensive evaluation of segmentation accuracy and robustness.

Table 3: Comparison of DGT-deeplabv3+ with other network models.

Model	mIoU	mAcc	mDice	mPrecision	mRecall
DeepLabV3+	78.01	82.67	85.96	90.01	82.67
PSPNet	77.71	83.15	85.72	88.73	83.15
Segformer	64.66	67.65	72.88	82.65	67.65
BiSeNetv2	76.89	80.87	85.04	90.53	80.87
FPN	78.21	82.06	86.13	91.04	82.06
GCN	78	83.10	85.96	89.37	83.10
DGT-deeplabv3+(ours)	**81.09**	**86.09**	**88.38**	**91**	**81.09**

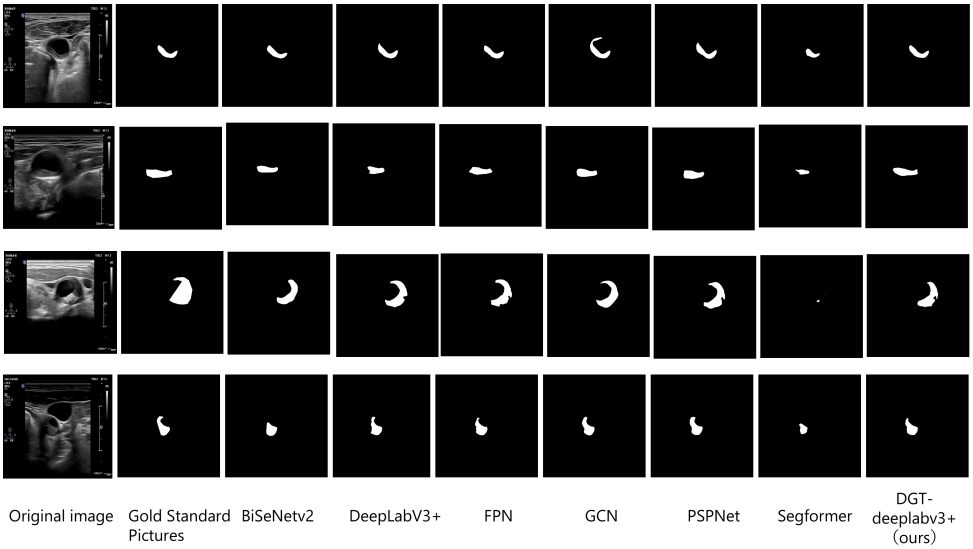

Figure 6: Comparison between DGT-deeplabv3+ and other 6 network structures to compare the effect of.

The results, shown in Table 3, indicate that while DeepLabV3+, PSPNet, and FPN perform well on large targets, Segformer, as a lightweight network, shows weaker performance in medical image segmentation, especially for small targets. In contrast, DGT-DeepLabV3+ significantly improves small target detection and complex boundary handling by leveraging DenseNet201 and MogaBlock for multi-scale feature extraction, the Grapher module for global context capture, and TverskyLDAMLoss for addressing class imbalance.

Specifically, DGT-DeepLabV3+ achieves an mIoU of 81.09%, 3.08% higher than DeepLabV3+ and 2.88% higher than FPN. It also shows a 3.42% improvement in mAcc and a 2.42% improvement in mDice, demonstrating superior performance in handling complex plaque shapes and edges.

Overall, DGT-DeepLabV3+ excels in small target detection and mitigating edge blurring. It effectively reduces over-segmentation in complex backgrounds and enhances robustness and accuracy through the integration of GCN and TverskyLDAMLoss.

Figure 6 presents the segmentation results of different methods on the experimental dataset, focusing on four images of irregularly shaped carotid plaques. The comparison highlights notable differences in plaque boundary recognition and detail handling across models.In the first image, BiSeNetv2, GCN, and PSPNet produce blurred boundaries, with Segformer and GCN failing to capture the complex shapes accurately due to overly smooth edges. In contrast, DGT-DeepLabV3+ provides clearer and more precise plaque contours.In the second image, U2Net shows slight oversegmentation, while DeepLabV3+ and FPN perform better but still miss some details. The proposed model excels here, addressing edge blurring effectively through the Grapher module and TverskyLDAMLoss, producing segmented plaques closer to the actual contours.In the third image, PSPNet and Segformer suffer from significant oversegmentation on small targets, while GCN and BiSeNetv2 fail to capture the plaque's full shape. DGT-DeepLabV3+, with its global feature extraction, achieves more accurate small plaque segmentation.Finally, in the fourth image, Segformer and PSPNet struggle with boundary ambiguity, and GCN and BiSeNetv2 show clear gaps in the segmentation. The proposed model delivers more stable boundary detection and clearer contours, avoiding oversegmentation and improving boundary precision.Overall, despite the complexity and irregularity of the plaques, the comparison in Figure 6 demonstrates that DGT-DeepLabV3+ excels in segmentation accuracy, edge clarity, and adaptability to complex shapes, showing clear advantages over other mainstream methods.

4 CONCLUSION

This study presents DGT-DeepLabV3+, an improved model based on DeepLabV3+, to address challenges like noise, low contrast, and blurry edges in carotid plaque segmentation. By using DenseNet201 as the backbone, the model reduces parameters while maintaining strong feature extraction, balancing accuracy and speed. MogaBlock enhances multi-scale feature extraction, and the integration of GCN into the ASPP module improves the handling of complex shapes and blurred boundaries.Though the inclusion of MogaBlock and GCN increases model complexity compared to the original DeepLabV3+, it significantly boosts segmentation accuracy, particularly for small and complex plaques. DenseNet201 reduces parameter count, balancing higher accuracy with reasonable computational cost. Future optimizations, such as model pruning, quantization, or hardware acceleration, could enhance real-time performance.The TverskyLDAMLoss function addresses class imbalance and improves small plaque segmentation. Further tuning of the Tversky and LDAM parameters could enhance performance on more complex, imbalanced datasets.While the model is validated for carotid plaque segmentation, further testing on other medical images is needed to assess generalizability. Transfer learning and training on larger datasets could improve robustness. For real-time applications, model compression and hardware efficiency are crucial to optimizing speed and practicality.In conclusion, DGT-DeepLabV3+ offers a promising solution for carotid plaque segmentation with potential for broader medical image analysis, pending further optimization and validation.

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